



A CuZn-BTC derived stable Cu/ZnO@SiO₂ catalyst for ethanol dehydrogenation

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ABSTRACT

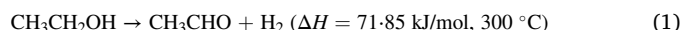
Cu-based catalysts play pivotal role in many industrial processes, however, its application under high temperature is restricted due to the low Tammann temperature of Cu. Improving the stability and activity of Cu-based catalysts at high temperature is a challenge. In this work, a stable Cu/ZnO@SiO₂ catalyst was prepared, in which silicon was introduced into CuZn-BTC precursor via a steam-assisted hydrolysis method before calcination and reduction. Characterizations found that Cu/ZnO@SiO₂ catalyst prepared via above strategy possesses increased surface area, more exposed active sites and excellent stability for ethanol dehydrogenation than Cu/ZnO (prepared from CuZn-BTC without SiO₂). It was confirmed that the space-time yield of acetaldehyde reached 9.0 g-acetaldehyde/g-cat/h, and Cu/ZnO@SiO₂ could maintain its activity within 200 h at 350 °C due to the ZnO decorated Cu particles (Cu/ZnO) were encapsulated in the octahedral shaped SiO₂ hollow.

1. Introduction

Ethanol is an important feedstock for its huge amount of productivity, renewable and low price. It was estimated that the productivity of bio-ethanol has reached 8×10^{10} liters in 2017, and catalytic upgrading of ethanol to valuable chemicals became a hot topic in the past decade [1–3]. Among of them, dehydrogenation of ethanol to acetaldehyde is an industrially interesting process that attracts the attention of many scientists [4–6], which might be attributed to that acetaldehyde is an important intermediate and is currently manufactured via environmental unfriendly Wacker-Hoechst process in the presence of a homogeneous PdCl₂/CuCl₂ catalyst in liquid-phase [7,8]. At the same time, the only by-product in the dehydrogenation of ethanol to acetaldehyde is hydrogen, which will make this process green, economical and sustainable.

In the past decade, Cu-based catalysts were recommended popularly for the dehydrogenation of ethanol for its high activity, selectivity and low price [9–13]. These works provided very important founding and insights on the reaction mechanism and structure-performance relationship of Cu-based catalysts for dehydrogenation of ethanol. However, dehydrogenation of ethanol is an endothermic process (see the following Eq. (1)) [13–15], and high temperature is more favorable to increase the

conversion of ethanol in single pass.



Unfortunately, Cu nanoparticles (NPs) agglomerate easily at high temperature due to the low Tammann temperature of Cu (405 °C) [16, 17]. Improving the stability and activity of Cu-based catalysts is a challenge for those reactions which might be carried out at high temperature. In the published works, several strategies have been developed in order to retard the growth of Cu NPs during the dehydrogenation of ethanol, these methods include using specific support (e.g. carbon materials [18], ZnO [19], Cr₂O₃ [20]), or alloying Cu with refractory metals (e.g. Ni [21], Pt [22]), or immobilization Cu in framework of hierarchical MFI [23] or encapsulated Cu within silicalite-1 [24]. And these approaches have made significant progresses, however, the stability and activity of Cu-based catalysts at high temperature (over 300 °C) remains a challenge.

Metal-organic frameworks (MOFs) possess highly ordered porous structure, uniform pore channels and high surface areas [25,26]. Metal ions that locate at the nodes of MOF dispersed homogenously and can work in synergy with encapsulated NPs [27–30]. At the same time, MOFs can also be used as self-sacrificing templates toward the

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fabrication of special structured catalysts. It was confirmed that MOF-derived metal NPs, alloy and/or oxides benefit from the precise morphologies and hierarchical porosities, and serve as excellent and durable heterogeneous catalysts for a variety of reactions [31–34]. Previous works in our laboratory found that a Cu(Zn)-BTC precursor derived Cu/ZnO catalyst with nano sized ZnO dotted on Cu particles was extremely active and stable for hydrogenolysis of glycerol due to the enlarged interface between Cu and ZnO [31]. But the direct calcination of MOFs precursor is a violent exothermic process, and thus, the surface area of synthesized metal NPs and/or oxides is low because of sintering [31,33].

Here, a stable Cu-based catalyst with increased surface area was synthesized via a controlled decomposition and reduction of a silica encapsulated CuZn-BTC. The structure and properties of the precursor and catalyst were characterized with various techniques. The prominent performance of the catalyst under varied conditions and the stability at high temperature were evaluated in detail.

2. Experimental

2.1. Materials

$\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$, ZnO, N,N'-dimethylformamide (DMF) and ethanol were purchased from Sinopharm Chemical Reagent Co., Ltd (Shanghai, China). Tetraethyl orthosilicate (TEOS) was obtained from Aladdin Biochemical Technology Co., Ltd (Shanghai, China). 1,3,5-benzenetricarboxylic acid (H_3BTC) was purchased from Energy chemical. All the reagents and solvents were used as received without further purification.

2.2. Catalysts preparation

2.2.1. Preparation of CuZn-BTC

CuZn-BTC was prepared in a rapid method at room temperature according to reference [35]. And the procedures were addressed in [Supporting information](#) section in detail.

2.2.2. Preparation of $\text{CuZn}^{\text{BTC}}\text{@SiO}_2$

SiO_2 encapsulated CuZn-BTC was synthesized via a steam-assisted hydrolysis process. At first, the as-synthesized CuZn-BTC (2.4 g) was degassed at 100 °C in vacuum ($< 10^{-7}$ Pa) for 6 h. And then, the system in vacuum was cooled to room temperature, and 15 mL TEOS was added to above pretreated CuZn-BTC. After that, the mixture was sonicated for 15 min and maintained overnight at room temperature in order to make the pore channels of CuZn-BTC full of TEOS, the precursor sediment was filtered. Subsequently, the solid was transferred to a small beaker and the beaker was placed in a Teflon-lined autoclave with 20 mL H_2O (see [Fig. S1](#)). The steam-assisted hydrolysis was performed at 110 °C for 12 h. Finally, the solid was dried at 80 °C in a vacuum for overnight and denoted as $\text{CuZn}^{\text{BTC}}\text{@SiO}_2$.

2.2.3. Preparation of CuZnO@SiO_2

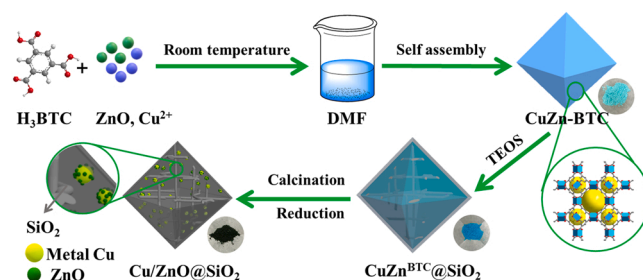
CuZnO@SiO_2 was prepared via the controlled calcination of above prepared $\text{CuZn}^{\text{BTC}}\text{@SiO}_2$ in air flow (30 mL/min) at 400 °C for 4 h with a ramp of 2 °C/min.

2.2.4. Preparation of Cu/ZnO@SiO_2

Before catalytic reaction, the calcined CuZnO@SiO_2 was pressed into granular (40–60 mesh), and then reduced at 350 °C in 5% H_2/N_2 flow (60 mL/min) for 1 h. The reduced catalyst was denoted as Cu/ZnO@SiO_2 . These procedures were diagrammatically depicted in [Scheme 1](#).

2.2.5. Preparation of reference catalysts

Cu/ZnO catalyst without SiO_2 hollow was prepared via the direct calcination and reduction of CuZn-BTC according to the reference [35], and these procedures were addressed in [Supporting information](#) section.



Scheme 1. Schematic illustration of the preparation of Cu/ZnO@SiO_2 .

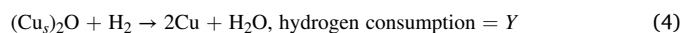
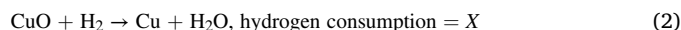
As references, Cu@SiO_2 and ZnO@SiO_2 catalysts were prepared in the same method, and preparation procedures for these catalysts were described in detail in [Supporting information](#).

2.3. Characterizations

X-ray diffraction (XRD) patterns were recorded on a Rigaku SmartLab diffractometer using $\text{Cu K}\alpha$ radiation ($\lambda = 1.5406 \text{ \AA}$) at 40 kV and 180 mA. N_2 adsorption-desorption isotherms were measured using an ASAP 2010 analyzer (Micromeritics) after degasification at 100 °C for 6 h in vacuum. The Brunner-Emmet-Teller (BET) surface area was calculated according to the adsorption branch in the range of $p/p_0 = 0.005\text{--}0.1$ (for CuZn-BTC and $\text{CuZn}^{\text{BTC}}\text{@SiO}_2$) or $0.05\text{--}0.30$ (for other samples). Thermogravimetric and differential scanning calorimeter (TG-DSC) analysis were performed on a Netzsch STA 409 thermoanalyzer with a heating rate of 2 °C/min in air flow (30 mL/min) from room temperature to 500 °C. Fourier-transfer infrared spectroscopic (FT-IR) analysis was carried on a Nicolet Is10 FTIR spectrometer using KBr method in $400\text{--}4000 \text{ cm}^{-1}$. X-ray photoelectron spectroscopy (XPS) spectra were detected on a Kratos Axis Ultra DLD system, and the X-ray source was standard Mg anode (1253.6 eV). Scanning electron microscopy (SEM) images were obtained on Hitachi SU8010 at 5 kV, and samples were sprayed with platinum in order to avoid charging. Transmission electron microscope (TEM) images and high angle angular dark field-scanning transmission electron microscopy (HAADF-STEM) images of these catalysts were obtained on JEOL-2020F at an accelerating voltage of 200 kV. Line-scan elemental energy-dispersive spectrometry (EDS) analysis was carried out with an FEI TECNAI F30 microscope at 200 kV.

H_2 temperature-programmed reduction (H_2 -TPR) experiments were carried out in the following procedures: samples (30 mg) were first pretreated at 450 °C for 1 h in Ar flow (30 mL/min) and cooled to 50 °C. Then, 10 % H_2/N_2 mixture was shifted to the reactor and the reactor was heated to 450 °C with a heating rate of 10 °C/min. The effluent gas was monitored with an online quadrupole mass spectrometer (OmniStarTM, GSD301, Switzerland).

N_2O oxidation and followed H_2 titration was carried out in the following procedure, (i) calcined samples were first reduced in the same procedures as above H_2 -TPR and the consumption of hydrogen (denoted as X) was recorded. (ii) The reactor was then cooled to 30 °C in purified Ar. (iii) 20% $\text{N}_2\text{O}/\text{Ar}$ was introduced to the catalyst for 60 min in order to make all surface Cu (denoted as Cu_s) oxidized. (iv) The reactor was purged with Ar for 120 min in order to remove all N_2O in reactor. (v) Another H_2 -TPR process was performed again, and the consumption of hydrogen was denoted as Y.



The dispersion of Cu was calculated as:

$$D = (2 \times Y/X) \times 100 \% \quad (5)$$

The area of surface Cu_2 was calculated as:

$$S = 2 \times Y \times N_{\text{av}} / (X \times M_{\text{Cu}} \times 1.4 \times 10^{19}) = 1353 \times Y / X \text{ (m}^2\text{-Cu/g-Cu)} \quad (6)$$

N_{av} is the Avogadro's constant and M_{Cu} is the relative atomic mass of copper (63.5 g/mol) in these equations.

2.4. Catalytic reaction

Catalytic reaction was performed in a vertical fixed-bed reactor at atmospheric pressure. Catalyst (0.3 g, sized in 40–60 mesh) was packed in the middle of quartz tube reactor ($I.D.$ = 8 mm, L = 270 mm), both sides of the reactor were sealed with quartz sand. Before reaction, catalyst was reduced at 350 °C for 1 h with 5% H_2/N_2 (60 mL/min), and it was cooled to desired temperature. Ethanol was fed into the reactor continuously using a high-pressure pump (Shimadzu, LC-20AD) with 20 mL/min purified nitrogen (99.99%, as the carrier gas and internal standard). The effluent gas from the reactor was cooled in a cold trap (kept at -25 °C), and the liquid products were selected, weighted and analyzed with a gas chromatograph (GC, Shimadzu, 14B) equipped with a flame ionization detector and a 30 m capillary column (DB-WAX 52 CB, USA). The gaseous products (such as H_2) were detected using an online GC (Schimadzu, 8 A) with a TCD detector and a Porapack Q column. The conversion of ethanol and the selectivity of each product were calculated based on the following equations:

$$\text{Conv.}(\%) = (N_{\text{in(ethanol)}} - N_{\text{out(ethanol)}}) / N_{\text{in(ethanol)}} \times 100\% \quad (7)$$

$$S_{\text{AcH}}(\%) = N_{\text{(AcH)}} / (N_{\text{in(ethanol)}} - N_{\text{out(ethanol)}}) \times 100\% \quad (8)$$

3. Results and discussion

3.1. The properties of CuZn-BTC and $\text{CuZn}^{\text{BTC}}\text{@SiO}_2$ precursors

XRD patterns of CuZn-BTC and $\text{CuZn}^{\text{BTC}}\text{@SiO}_2$ precursors were presented in Fig. 1(a). It was confirmed that well-structured $(\text{CuZn})_3(\text{BTC})_2$ formed in CuZn-BTC , and it exhibited the same diffraction patterns as the typical $\text{Cu}_3(\text{BTC})_2$ (JCPDS 00–064–0936) because of the similar radius of Cu^{2+} (7.2 Å) and Zn^{2+} (7.4 Å) [36]. And it was interesting to note that the intensity of these diffraction peaks of $\text{CuZn}^{\text{BTC}}\text{@SiO}_2$ was lower, which might be attributed to the accumulation of SiO_2 on CuZn-BTC . But the primary profiles and the relative intensities of these diffraction peaks of $\text{CuZn}^{\text{BTC}}\text{@SiO}_2$ were similar as that of $\text{Cu}_3(\text{BTC})_2$, which manifested that the framework of CuZn-BTC remained even after the introduction of SiO_2 . At the same time, the calculated cell parameters of CuZn-BTC and $\text{CuZn}^{\text{BTC}}\text{@SiO}_2$ ($a=b=c=2.644$ nm, see Table S1) fitted well with the data of typical $\text{Cu}_3(\text{BTC})_2$ that reported in literature (2.643 nm) [37].

FTIR spectra of CuZn-BTC and $\text{CuZn}^{\text{BTC}}\text{@SiO}_2$ were compared in Fig. 1(b). The C=C stretching in benzene ring (1375 cm^{-1}), the C=O stretching vibration of trimesic acid (1647 cm^{-1}) and the scissoring vibration of the carboxylate ion (730 cm^{-1}) appeared clearly and fitted well with the previously reported $\text{Cu}_3(\text{BTC})_2$ structure in literature [38]. Apart from the typical signals of $\text{Cu}_3(\text{BTC})_2$, another three peaks appeared at 1095, 934 and 870 cm^{-1} in $\text{CuZn}^{\text{BTC}}\text{@SiO}_2$, which were attributed to the characteristic vibrations of Si-O-Si groups [39]. These results confirmed that SiO_2 was successfully introduced in the framework of CuZn-BTC .

N_2 adsorption isotherms of both samples were classical type I model with obvious N_2 uptake in the range of 0.0–0.03 of p/p_0 (see Fig. 1(c)), indicating the presence of micropores. Fig. 1(d) further confirmed that CuZn-BTC possesses double microporous pores that sized in 0.73 and

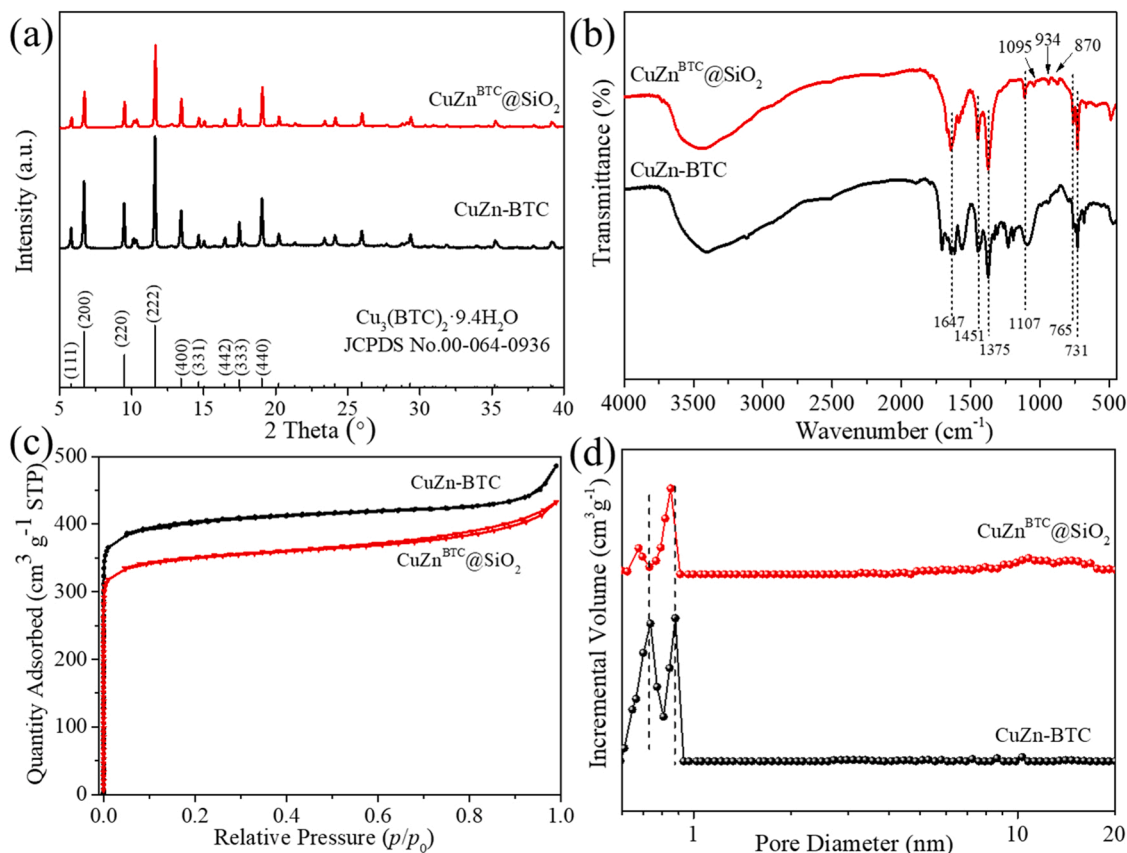


Fig. 1. XRD (a), FTIR (b), N_2 adsorption-desorption isotherms (c) and pore size distribution (d) of CuZn-BTC and $\text{CuZn}^{\text{BTC}}\text{@SiO}_2$.

0.87 nm, respectively, which was consistent with previous literatures [40]. On the other hand, the diameter of these pore channels decreased to 0.67 and 0.83 nm when SiO₂ was introduced in CuZn^{BTC}@SiO₂. This decreased diameter might be attributed to that SiO₂ would penetrate the pore channels of CuZn-BTC. The calculated surface area and pore volume of CuZn^{BTC}@SiO₂ also decreased from 1606 m²/g and 0.55 cm³/g (of CuZn-BTC) to 1202 m²/g and 0.42 cm³/g, respectively (see Table S1). These results further indicated that SiO₂ was effectively introduced into the channels and/or the surface of CuZn-BTC.

Fig. 2 compared the TG analysis results of CuZn-BTC and CuZn^{BTC}@SiO₂ in air flow. It was found that there were three mass loss stages during the calcination of both samples. The first stage at 25–150 °C was ascribed to the volatilization of adsorbed water or solvent, and the second mass loss between 150 and 300 °C might result from the decomposition of organic linkers at the edge of CuZn-BTC. After that, organic linkers in CuZn-BTC began to burning violently accompanied with two huge exothermic peaks at around 318.4 and 333.8 °C [41]. Compared with CuZn-BTC, the burning of organic linkers in CuZn^{BTC}@SiO₂ delayed to 358.4 °C and the decomposition process became gentler in the temperature ranged from 320 to 365 °C. These results indicated that the introduction of SiO₂ (or the coating of SiO₂ outside CuZn-BTC) could prevent the violent burning of organic linkers. That is, the stability of the CuZn-BTC could be improved after the introduction of SiO₂. According to the total mass loss before 500 °C, the calculated mass ratio of SiO₂ in the final CuZnO@SiO₂ is 23.8 wt%.

Fig. 3 exhibited the typical SEM images and surface morphologies of CuZn-BTC and CuZn^{BTC}@SiO₂. It can be found that pristine CuZn-BTC had a typical regular octahedral profile with a clear surface, and the corresponding EDS analysis confirmed Cu and Zn dispersed uniformly in the particles. It is interesting to find that the octahedral outline remained well in CuZn^{BTC}@SiO₂, but the surface of these octahedrons became rough because of the introduction of SiO₂, which made the clear CuZn-BTC particles covered. EDS analysis further disclosed that the content of Si, Cu and Zn changed simultaneously in the octahedral CuZn^{BTC}@SiO₂ particles.

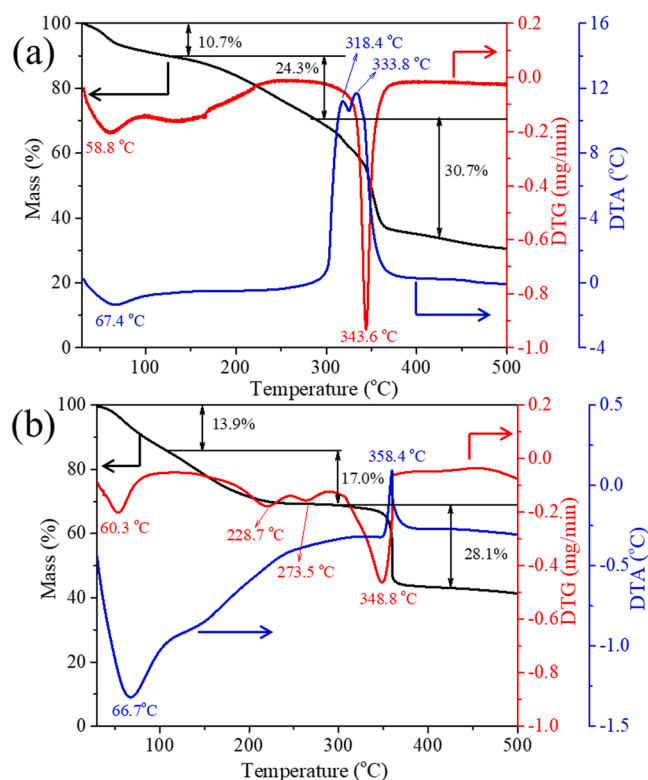


Fig. 2. Thermal analysis of CuZn-BTC (a) and CuZn^{BTC}@SiO₂ (b).

3.2. The properties of calcined CuZnO@SiO₂

Fig. 4(a) showed the XRD patterns of calcined CuZnO@SiO₂ and compared with the reference CuZnO that prepared via the direct calcination of CuZn-BTC (without SiO₂). Obvious CuO phase (JCPDS 00-048-1548) with three typical peaks at $2\theta = 43.4^\circ$, 50.4° and 74.1° was detected in both samples. Meanwhile, the broad peak in $19-25^\circ$ in CuZnO@SiO₂ might be attributed to the signal of amorphous silica. But it was interesting to note that the characteristic reflections of CuO in CuZnO@SiO₂ became weaker and the calculated crystalline size of CuO also decreased from 14.6 nm (in CuZnO) to 3.9 nm. These results suggested that the introduction of SiO₂ could make the burning of organic linkers gentler (see Fig. 2) and depress the sintering of CuO particles during the calcination process.

At the same time, N₂ adsorption-desorption analysis further disclosed that the surface area of CuZnO@SiO₂ reached 139.7 m²/g, which is 25 times higher than that of CuZnO (5.3 m²/g) (see Fig. 4(b) and Table 1). The pore volume of CuZnO@SiO₂ reached 0.33 cm³/g which is higher than that of CuZnO (0.02 cm³/g). Once again, these data inferred that SiO₂ could prevent the violently collapse of CuZn-BTC's framework and depress the sintering of CuO and/or ZnO during calcination.

Fig. S2 compared H₂-TPR profiles of calcined CuZnO and CuZnO@SiO₂. There was only one reduction peak in both samples, which could be attributed to the reduction of Cu²⁺ to Cu⁰ and/or Cu⁺ [42]. But the reduction temperature increased obviously from 215 °C (in CuZnO) to 245 °C (in CuZnO@SiO₂) when SiO₂ was introduced, indicating that CuO species in CuZnO@SiO₂ were more stable.

N₂O titration and followed H₂ reduction analysis further disclosed that the dispersion of Cu in CuZnO@SiO₂ reached to 20.4%, which was several times higher than that of CuZnO (3.5%, without SiO₂) that was prepared via the direct calcination of pristine CuZn-BTC [35]. Meanwhile, the calculated average diameter of Cu in reduced CuZnO@SiO₂ and CuZnO was 4.9 and 22.2 nm, respectively. The higher dispersion of Cu in CuZnO@SiO₂ might be attributed to that SiO₂ prevented the sintering of CuO and the larger surface area of CuZnO@SiO₂ (see Fig. 4).

XPS survey spectra of CuZnO and CuZnO@SiO₂ was shown in Fig. S3, and the obvious signals of Si 2p were detected in CuZnO@SiO₂. At the same time, high-resolved Cu 2p spectrum (Fig. 5) indicated that the binding energy of Cu 2p_{3/2} and Cu 2p_{1/2} shifted from 933.9 and 953.6 eV (in CuZnO) to 934.4 and 954.2 eV (in CuZnO@SiO₂), indicating that the interaction between Cu and ZnO (and/or between Cu and SiO₂) became stronger after the introduction of SiO₂. The binding energy at 942–944 eV in the XPS spectrum is attributed to the satellite of Cu²⁺. After reduction, the overall binding energy of Cu 2p_{3/2} and Cu 2p_{1/2} decreased to 932.9 and 952.8 eV, respectively, because of the formation metallic Cu and/or Cu⁺. The existence of Cu⁰ in reduced Cu/ZnO@SiO₂ was also confirmed via Cu LMM auger spectra (see Fig. S4). The calculated mol ratio of Cu⁰ to Cu⁺ in Cu/ZnO@SiO₂ was 1.25, which might be attributed to the strong interaction between Cu and SiO₂. And the synergism between Cu⁺ and Cu⁰ would improve the activity of Cu/ZnO@SiO₂ [23]. It is worth noting that the surface content of Cu in CuZnO@SiO₂ was lower than the bulk content (see Table 1), which might be attributed to that Cu and/or ZnO species was covered by the added SiO₂.

3.3. The properties of reduced Cu/ZnO@SiO₂

Fig. S5 displayed the XRD pattern of reduced Cu/ZnO@SiO₂ and compared with the reference Cu/ZnO (without SiO₂). It was found that mainly metallic Cu (JCPDS 01-085-1326) and ZnO (JCPDS 00-065-0725) were detected in both samples. At the same time, the calculated crystalline size of Cu in these catalysts was 4.0 nm and 22.0 nm, respectively, which was consistent with the above N₂O oxidation. These results suggested that the introduction of SiO₂ could improve the dispersion of Cu and inhibit the agglomeration of Cu during the reduction process.

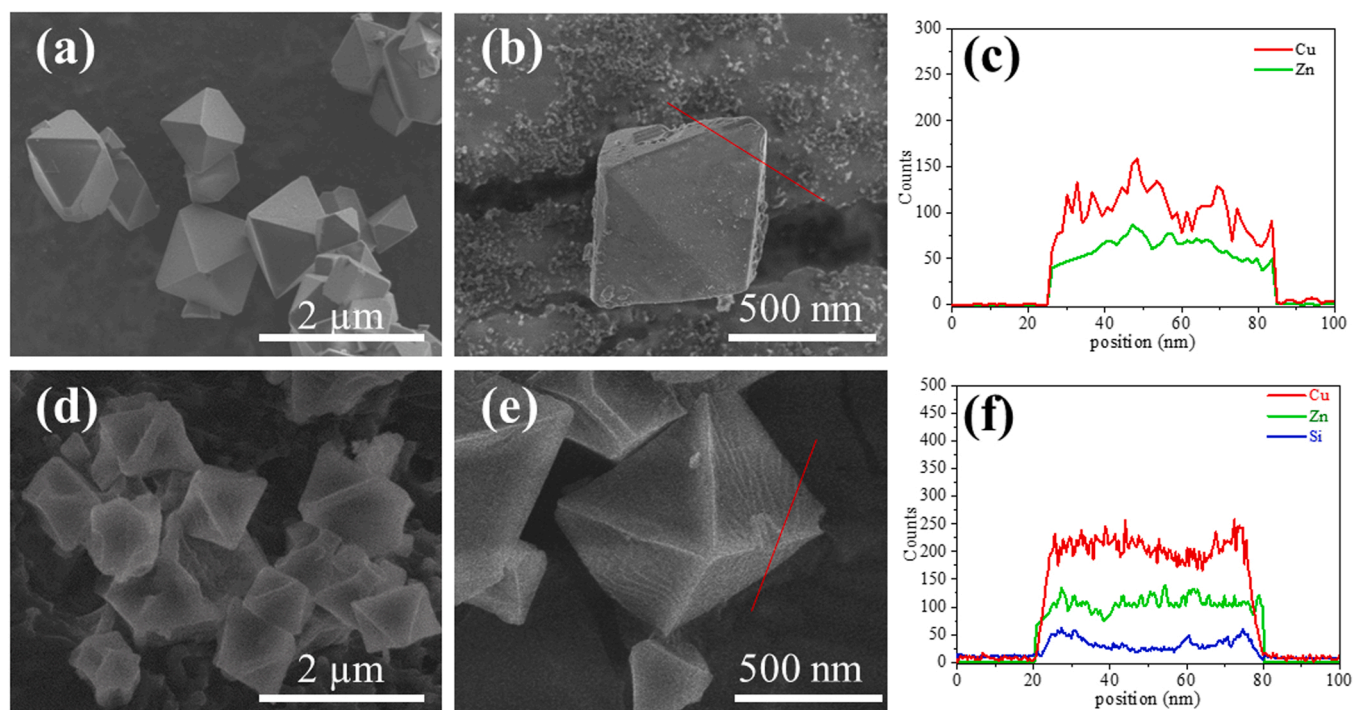


Fig. 3. SEM images and corresponding line-EDS of CuZn-BTC (a-c) and CuZn^{BTC}@SiO₂ (d-f).

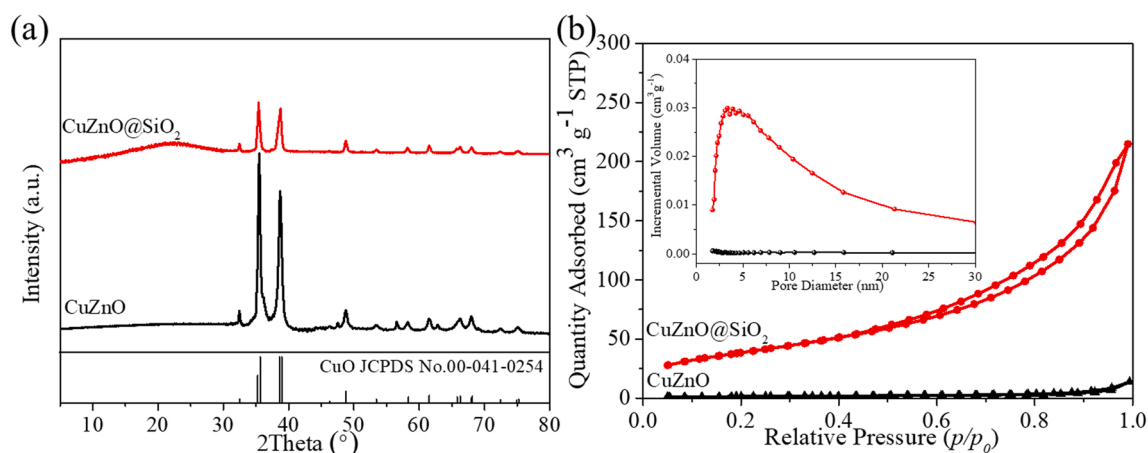


Fig. 4. XRD (a), N₂ adsorption-desorption isotherms (b) and pore size distribution (insert) of CuZnO and CuZnO@SiO₂.

Table 1

Textural properties of CuZnO and CuZnO@SiO₂.

Sample	S _{BET} (m ² /g)	Pore volume (cm ³ /g) ^a	Cu dispersion (%) ^b	Cu particle size (nm) ^b	Surface content ^c (%)	Bulk content ^d (%)
CuZnO	5.3	0.02	3.5	22.2	45.5	39.6
CuZnO@SiO ₂	139.7	0.33	20.4	4.9	11.8	30.0
CuZnO@SiO ₂ ^e	112.4	0.30	17.0	5.9	10.6	30.1

^a Calculated in Barrett-Joyner-Halenda method.

^b Calculated from N₂O titration (at 30 °C) after the catalysts were reduced in H₂ at 350 °C.

^c Data from XPS analysis.

^d Data from ICP analysis.

^e Spent CuZnO@SiO₂ after time on stream for 200 h at 350 °C.

Fig. 6(a) revealed the TEM images of Cu/ZnO@SiO₂, which confirmed the octahedral profile was still maintained in reduced Cu/ZnO@SiO₂. Meanwhile, both the elemental mapping of a separated octahedral particle (see Fig. 6(a)) and the line-scanning elemental

analysis of enlarged images (see Fig. 6(b) and (d)) indicated that an octahedral shaped SiO₂ hollow formed in Cu/ZnO@SiO₂ catalyst, and most Cu/ZnO particles were encapsulated in SiO₂ hollow. More interestingly, HAADF image and corresponding line-scanning elemental

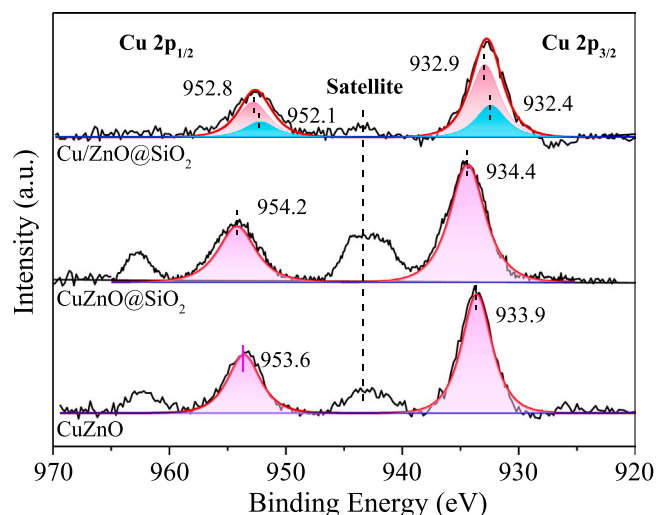


Fig. 5. Binding energies of Cu 2p in CuZnO, CuZnO@SiO₂ and Cu/ZnO@SiO₂.

analysis (see Fig. 6(d)), the fast Fourier transformation (FFT) analysis of HRTEM images with the lattice plane intensity profile (see Fig. 6(e)–(g)) disclosed that those reduced Cu particles (sized in 4–5 nm) were dotted with even smaller ZnO particles (sized in 1–2 nm). The interplanar spacing of the pure copper phase is 0.258 nm (110) (see Fig. 6(e)–(f)), and the interplanar spacing of ZnO is 0.241 nm (101) [21,43] (see Fig. 6(e)–(g)).

3.4. Catalytic performance

Table S2 compared the performance of reference ZnO@SiO₂ and Cu@SiO₂ catalysts for the dehydrogenation of ethanol in a vertical fixed-bed reactor at 310 °C. It was found that (1) ZnO@SiO₂ was inactive for this reaction, (2) Cu@SiO₂ was active for the dehydrogenation with a moderate conversion of ethanol (40.1%) and (3) the conversion of ethanol reached 84.2% over Cu/ZnO@SiO₂ under the same reaction conditions. These results confirmed that Cu was indispensable for the dehydrogenation of ethanol. At the same time, the higher activity of Cu/ZnO@SiO₂ than that of Cu@SiO₂ further indicated that ZnO could improve the activity of Cu for the dehydrogenation of ethanol. The promotion effect of ZnO on the performance of Cu-based catalysts were also reported in methanol synthesis [44–46], methanol decomposition, steam reforming and partial oxidation to hydrogen [47].

Table 2 summarized the initial performance of Cu/ZnO@SiO₂ and Cu/ZnO (prepared via the direct calcination and reduction of pristine CuZn-BTC without SiO₂) for the continuous dehydrogenation of ethanol at 310 °C. It was found that the conversion of ethanol over Cu/ZnO was 75.9%, and selectivity of acetaldehyde reached 98.2%. Under the same conditions, Cu/ZnO@SiO₂ exhibited better performance than that of Cu/ZnO, and the detected conversion of ethanol and selectivity of acetaldehyde increased to 84.2% and 99.1%, respectively. The calculated space time yield (STY) of acetaldehyde over Cu/ZnO@SiO₂ reached 8.1 g-AcH/g-cat/h at 310 °C. This improvement might benefit from the high specific surface area and high dispersion of copper in Cu/ZnO@SiO₂. In addition, Fig. S6 compared the stability of Cu/ZnO and Cu/ZnO@SiO₂ for the dehydrogenation of ethanol at 310 °C. It was found that the Cu/ZnO deactivated quickly as the detected conversion of ethanol decreased sharply to 40.1% (within 40 h). On the other hand, it was interesting to note that Cu/ZnO@SiO₂ could maintain its activity for 50 h, mainly thanks to that Cu/ZnO were further encapsulated in SiO₂ hollow. The advantages of SiO₂ modified catalyst could be attributed to its high surface area and stabilized catalyst structure.

The performance of Cu/ZnO@SiO₂ for dehydrogenation of ethanol under varied reaction temperature (250–390 °C) was shown in Fig. 7. It

was found the conversion of ethanol was limited (50.6%) at lower temperature (250 °C) and it increased quickly to 84.2% (at 310 °C), after that, the conversion of ethanol increased slowly to 91.2% (at 350 °C) and 93.8% (at 390 °C). On the other hand, the selectivity of acetaldehyde decreased slightly from 99.6% (at 250 °C) to 98.0% (at 390 °C), which might be attributed to the highly reactivity of acetaldehyde [48].

Fig. S7 exhibited the performance of Cu/ZnO@SiO₂ for the dehydrogenation of ethanol under different contact time at 310 °C. It was found that the detected conversion of ethanol increased rapidly from 30.9% (at 0.06 h) to 84.2% (at 0.32 h), and remained higher than 85% under the prolonged contact time. However, the selectivity for acetaldehyde decreased slightly under longer contact time due to the reactivity of acetaldehyde.

The stability of Cu/ZnO@SiO₂ catalyst for the dehydrogenation of ethanol was checked under extremely harsh condition (350 °C) (see Fig. 8). At first, it was confirmed Cu/ZnO@SiO₂ was stable at 310 °C within 50 h as the detected conversion of ethanol changed slightly around 84%. After that, the conversion of ethanol could be increased to 88% and remained stable within another 50 h when the temperature was raised to 330 °C. Finally, the temperature was increased to 350 °C in order to accelerate the deactivation of Cu/ZnO@SiO₂. It was quite interesting to note that the detected conversion of ethanol only decreased slightly from 91.2% to 90.1% within the followed 200 h. And the selectivity of acetaldehyde remained higher than 98.5% within all these 300 h. To the best of our knowledge, this prominent stability of Cu-based catalyst at high temperature (350 °C) for the dehydrogenation of ethanol was seldom reported in literatures. Characterization results indicated that Cu and ZnO particles still dispersed homogeneously in spent catalyst with a slightly increased size (see Figs. S8 and S9 and Table 1). But the octahedral profile collapsed partially (see Fig. S8), which might be attributed to the violent mechanical treatments of spent Cu/ZnO@SiO₂. These destructive treatments include tablet compressing (before reaction), scouring of hot gas (350 °C, 200 h) and grinding (before TEM analysis). According to above characterization results, the stability of Cu/ZnO@SiO₂ could be attributed to the following reasons: (1) Cu and Zn dispersed homogeneously in its precursor (CuZn^{BTC}@SiO₂, see Fig. 3b), and (2) those Cu particles (4–5 nm) were dotted with fine ZnO particles in the final catalyst which could make Cu particle isolated from each other (see Fig. 6(d)&(e)). At last, (3) SiO₂ hollow could also protect the sintering of these Cu/ZnO particles. These results confirmed that Cu/ZnO@SiO₂ is an excellent catalyst for the dehydrogenation of ethanol compared with those published works for its higher STY and stability (see Table S3 for details) [21,23,49–53].

4. Conclusions

In conclusion, a stable Cu-based catalyst was prepared via the controlled calcination and reduction of a silica encapsulated CuZn-BTC. Characterization results indicated that SiO₂ could prevent the violent burning of organic linkers during calcination, retard the sintering of Cu/ZnO particles. The final Cu/ZnO@SiO₂ catalyst possesses increased surface area, highly dispersed Cu particles and most Cu particles were dotted and isolated with fine ZnO. At the same time, SiO₂ hollow could also protect the sintering of these Cu/ZnO hybrids. As a result, Cu/ZnO@SiO₂ catalyst exhibited prominent activity and stability for the dehydrogenation of ethanol. The selectivity and STY of acetaldehyde reached 98.5% and 9.0 g-acetaldehyde/g-cat/h, respectively, at 350 °C, 0.1 MPa and WHSV = 3.1 h^{−1}. Meanwhile, Cu/ZnO@SiO₂ could maintain its activity within 200 h on stream in a severe condition (350 °C). This strategy will improve the lifetime of Cu-based catalysts under harsher conditions and broaden their application in the field of heterogeneous catalysis.

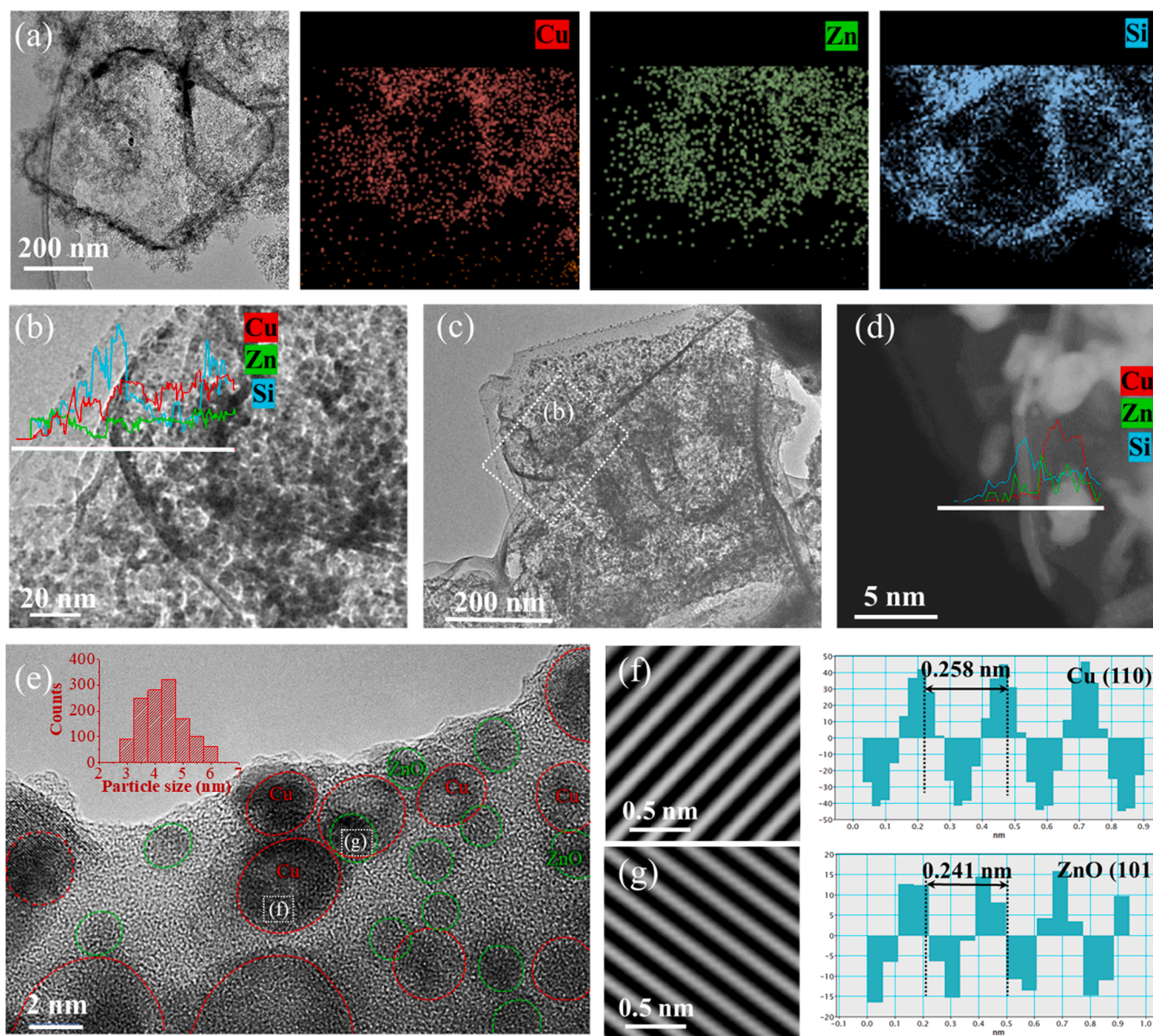


Fig. 6. HRTEM images (a, b, c, e) and the HAADF image (d) of Cu/ZnO@SiO₂; the fast Fourier transformation (FFT) of the HRTEM images (f, g) with the lattice plane intensity profile.

Table 2

Dehydrogenation of ethanol over Cu/ZnO and Cu/ZnO@SiO₂.

Catalyst	Conv. (%)	Selectivity (%)					STY (g-AcH/g-cat/h) ^a
		AcH ^b	EtOAc ^c	HOAc ^d	Et ₂ O ^e	Others ^f	
Cu/ZnO	75.9	98.2	0.4	0.2	0.6	0.6	7.5
Cu/ZnO@SiO ₂	84.2	99.1	0.3	–	0.4	0.2	8.1

Reaction conditions: catalyst 0.3 g; 310 °C; 0.1 MPa, 20 mL/min N₂; ethanol feed in 1.2 mL/h, WHSV = 3.2 h⁻¹.

^a STY, defined as (gram of formed acetaldehyde)/ (gram of catalyst)/ (contact time).

^b AcH, acetaldehyde.

^c EtOAc, ethyl acetate.

^d HOAc, acetic acid.

^e Et₂O, diethyl ether.

^f Others, mainly containing acetaldehyde oligomer.

CRediT authorship contribution statement

Haolan Liu: Investigation, Methodology, Formal analysis, Writing - Original Draft. **Zhili Chang:** Investigation, Validation, Formal analysis.

Jie Fu: Discussion, Supervision, Funding acquisition. **Zhaoyin Hou:** Conceptualization, Resources, Writing - Review & Editing, Supervision, Funding acquisition.

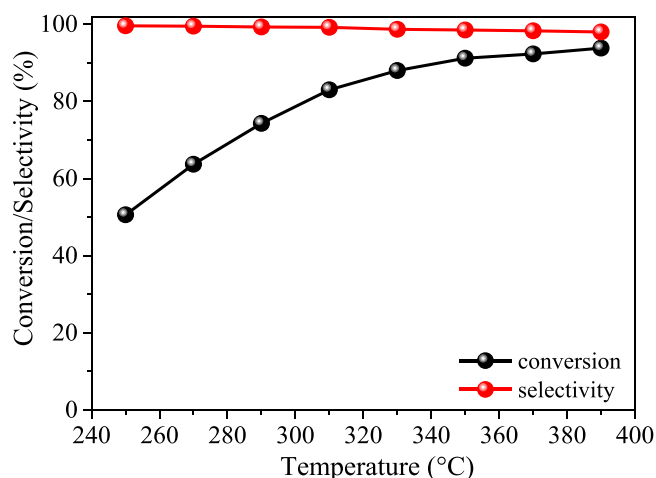


Fig. 7. Dehydrogenation of ethanol over Cu/ZnO@SiO₂ at varied temperature. Reaction conditions: catalyst 0.3 g; 0.1 MPa, 20 mL/min N₂; ethanol feed 1.2 mL/h, WHSV = 3.2 h⁻¹.

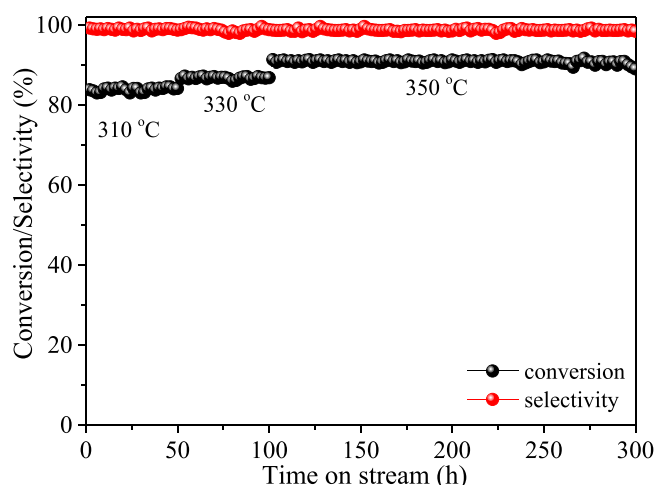


Fig. 8. Stability of Cu/ZnO@SiO₂ for the dehydrogenation of ethanol. Reaction conditions: catalyst 0.3 g; 0.1 MPa, 20 mL/min N₂; ethanol with feed of 1.2 mL/h, WHSV = 3.2 h⁻¹.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.apcatb.2022.122194](https://doi.org/10.1016/j.apcatb.2022.122194).

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